Borel subgroups and duality in coset twisting

Independent correction note for the N006 source contract

11 October 2026

Abstract

We check several literal statements in the 28 May 2026 arXiv revision of Piedade–Tranchida, *The Geometry of Wreath and Semi-Direct Products*. A thin rank-one wreath example gives the incidence graph $K_{2,8}$. Self-dual polytopes of type $\{2,2\}$ and $\{2,2,2\}$ disprove, respectively, the unqualified automorphism embedding and the universal nonlinear-diagram conclusion. We also derive the actual wreath parabolics, separate two missing hypotheses in the preliminary parabolic lemma, and exhibit an outer automorphism quotient with no homomorphic section. These are bounded corrections, not a review or recovery of the whole paper.

1 The conventions that matter

All page references refer to the fixed 22-page source [1]. Physical and printed page numbers agree. Its Section 2.1 allows a group action on a geometry through its image in the automorphism group (p. 3). Thus a flag-transitive coset model need not be faithful. Write

$$A_J = \bigcap_{j\in J} A_j, \qquad A_\varnothing = A.$$

In particular, $C = A_{I_\alpha}$ is the Borel subgroup. This intersection notation is different from the subgroup generated by a specified subset of distinguished generators. Definition 3.10 uses the intersection notation. The string C-group definition on p. 5 imposes commutation on nonconsecutive generators but does not require consecutive products to have order greater than two. We shall prove directly that the examples using such products are admissible abstract regular polytopes.

2 Wreath parabolics and thinness

Let B act on a finite set Ω. The product geometry has group $\bar{A} = \prod_{\omega\in\Omega} {}^{\omega}A$ and maximal parabolic at (j,ω) obtained by replacing the ω-factor by A_j. An orbit of types is $O \times \{j\}$, where O is a B-orbit on Ω. Choose a representative $\lambda_{O,j} \in O$ for each such orbit. For $i \in I_\beta$ put

$$J_i(\omega) = \{j \in I_\alpha \,:\, \omega \notin B_i\lambda_{O(\omega),j}\}.$$

Proposition 1. *The parabolic prescribed by Definition 3.10 for a type $i \in I_\beta$ is*

$$G_i = \left(\prod_{\omega\in\Omega} {}^{\omega}A_{J_i(\omega)}\right) \rtimes B_i. \tag{1}$$

If the representatives $\lambda_{O,j} = \lambda_O$ are independent of j, set $U_i = \bigcup_O B_i\lambda_O$. Then this becomes

$$G_i = \left(\prod_{\omega\in U_i} {}^{\omega}A \times \prod_{\omega\notin U_i} {}^{\omega}C\right) \rtimes B_i. \tag{2}$$

Proof. Definition 3.10 intersects the product parabolics indexed by the complement of each B_i-orbit of the chosen representative. At coordinate ω, these are exactly the restrictions A_j with $j \in J_i(\omega)$. Intersections in a direct product are coordinatewise, proving (1). With coherent representatives, $J_i(\omega)$ is empty inside U_i and is all of I_α outside it. This gives (2). Each indicated product is B_i-invariant, as required for the displayed semidirect product. □

Proposition 4.3 on p. 15 instead prints full A-factors on the complement of U_i, and does not retain the Borel factors. Even when $C = 1$, the support of the full factors should be U_i, not its complement. When representatives depend on j, formula (1) is needed. The worked symmetric-group example on p. 17 does use the support U_i, consistent with Definition 3.10 and (2).

Proposition 2. *The thinness conclusion of Theorem 4.4(d), under its literal stated hypotheses, fails for rank-one thin factors. There is a twisting with group order* 48 *and incidence graph* $K_{2,8}$.

Proof. Take $\alpha = (C_2, (1))$ and $\beta = (S_3, (C_3))$, with their single types. Each geometry consists of two elements, so it is thin and flag-transitive; rank-one residual connectedness is vacuous under the source definition. Let S_3 act naturally on $\Omega = \{1, 2, 3\}$. Both S_3 and its Borel C_3 are transitive on Ω. In the rank-one orbit-intersection condition, the two subsets of I_β therefore both yield Ω. All four required intersection equalities hold.

Now $\bar{A} = C_2^3$ and $G = C_2^3 \rtimes S_3$. There is one type orbit in the product. Definition 3.10 gives the two maximal parabolics

$$P = C_2^3 \rtimes C_3, \qquad Q = S_3, \qquad P \cap Q = C_3.$$

Indeed the B_i-orbit is all three coordinates, so its complement imposes no restrictions on $\bar{A}$; the parabolic for the type orbit imposes all three restrictions and leaves only Q. Thus $|P| = 24$, $|Q| = 6$ and $|P \cap Q| = 3$. The product PQ has 48 elements and equals G. Every left coset of P consequently meets every left coset of Q. There are two cosets of P and eight cosets of Q, proving the asserted incidence graph. The two panel sizes are $[P : P \cap Q] = 8$ and $[Q : P \cap Q] = 2$, so the geometry is not thin. □

Theorem 3.16(e) on p. 12 explicitly assumes trivial Borel subgroups for thinness, whereas Theorem 4.4(d) omits that assumption. Replacing β by its faithful two-point model does not preserve this example's action on Ω: the normal subgroup C_3 acts nontrivially there, so the action does not factor through S_3/C_3. The paragraph following Theorem 3.16 therefore does not repair this particular application. This does not disprove Corollary 3.17 when its geometries are modeled by their faithful automorphism groups; the interpretation of the acting models must be stated.

3 Self-dual polytopes with all adjacent labels two

Lemma 3. *For every $n \geq 2$ there is a self-dual abstract regular n-polytope with automorphism group $V = \mathbb{F}_2^n$, distinguished generators the coordinate vectors $e_0, \ldots, e_{n-1}$, and type $\{2, \ldots, 2\}$. Its associated coset geometry has two objects of each type, with every two objects of different types incident.*

Proof. Adjoin a least face of rank -1 and a greatest face of rank n to two faces $(i, 0), (i, 1)$ of each rank $0 \leq i < n$. Declare every face of lower rank smaller than every face of higher rank; the two faces at one rank are incomparable. Every maximal chain has one face of each rank. An interval whose endpoints differ by two ranks has exactly the two faces of the intermediate rank, so the diamond condition holds. A flag is a choice at some of the ranks. Its completing chambers are all independent choices at the remaining ranks, and any two are joined by changing one choice at a time. This also proves strong flag connectivity while keeping any common faces fixed, for the polytope and each of its sections.

A rank-preserving automorphism can independently swap the two faces of each proper rank, and has no other freedom. The full group is therefore V, acting regularly on the 2^n chambers. The map

$(i, \varepsilon) \mapsto (n-1-i, \varepsilon)$, interchanging least and greatest faces, is an order-reversing duality fixing the chamber of zero choices. It induces the involutory automorphism $\tau(e_i) = e_{n-1-i}$ of V.

For completeness, the subgroup generated by $\{e_i : i \in S\}$ consists exactly of vectors supported on S. Two such subgroups intersect in the one supported on the intersection of their supports. This proves the full C-group intersection property; all distinct generator products have order two, so the string relations hold. The cosets of $V_i = \langle e_j : j \neq i \rangle$ are specified by the ith coordinate. Different coordinate conditions are compatible, so the coset geometry is exactly the incidence geometry just described. It is thin, flag-transitive and residually connected: every residue of rank at least two is a complete multipartite graph with two vertices in each remaining part. □

This explicit construction verifies admissibility without assuming that the source uses a restrictive convention excluding digons. Its definitions on pp. 3 and 5–6 contain no such restriction.

Proposition 4. *The conclusion of Theorem 5.2(d) fails for the rank-three polytope of Lemma 3. More generally, if $\tau \in \mathrm{Aut}(A)$ is an outer involution, the canonical conjugation map has kernel $Z(A)$ and gives*

$$(A \rtimes \langle \tau \rangle)/Z(A) \;\cong\; \langle \mathrm{Inn}(A), \tau \rangle \;\leq\; \mathrm{Aut}(A).$$

Consequently $Z(A) = 1$ is a sufficient extra hypothesis for the claimed canonical embedding.

Proof. For $A = \mathbb{F}_2^3$, reversal swaps e_0, e_2 and fixes e_1. It is nontrivial, and hence outer because A is abelian. The semidirect product has order 16, whereas

$$|\mathrm{Aut}(A)| = |\mathrm{GL}_3(2)| = (8-1)(8-2)(8-4) = 168.$$

Since 16 does not divide 168, no embedding exists, even an unspecified one. In general the map $(a, \tau^\varepsilon) \mapsto \mathrm{inn}(a)\tau^\varepsilon$ is a homomorphism onto the displayed subgroup. An element of its kernel with $\varepsilon = 1$ would make τ inner, which is excluded. With $\varepsilon = 0$, the kernel condition is exactly $a \in Z(A)$. The center is τ-invariant, so the quotient is well-defined and the assertion follows. □

Proposition 5. *The nonlinear-diagram assertion in Theorem 5.2(b) fails for the rank-four polytope of Lemma 3. Its group $H = \mathbb{F}_2^4 \rtimes \langle \tau \rangle$ has no rank-three regular hypertope with a nonlinear diagram, irrespective of the chosen distinguished generators.*

Proof. Write elements as (v, ε), with multiplication

$$(v, \varepsilon)(w, \delta) = (v + \tau^\varepsilon w, \varepsilon + \delta).$$

All involutions in $V \times \{0\}$ commute. An element $(v, 1)$ is an involution exactly when $v + \tau v = 0$, that is, v is fixed by τ. Two such involutions commute, since their products in either order equal $(v + w, 0)$. Thus the involutions split into two sets, each with pairwise commuting elements. Every triple of involutions contains a commuting pair.

We recall why this is a geometric obstruction. In a thin residually connected flag-transitive geometry, the chamber graph is connected. This follows by induction on rank: chambers through a fixed object are connected by induction in its residue, and a path in the incidence graph joins these connected sets; rank one starts with its two chambers. A type-preserving automorphism fixing one chamber fixes each unique adjacent chamber by thinness, hence all chambers by connectedness, and therefore every object. The group consequently acts regularly on chambers. The unique automorphism sending a base chamber to its i-adjacent chamber is an involution ρ_i. These involutions generate the group because the chamber graph is connected. For two types i, j, the rank-two residue has chamber graph the alternating cycle generated by ρ_i, ρ_j; its gonality is the order of $\rho_i \rho_j$. In particular, commuting distinct generators give gonality two and no diagram edge.

In rank three, every diagram with at least one missing edge is linear after ordering its three vertices along a path. A nonlinear diagram therefore needs three pairwise noncommuting distinguished involutions, which H does not have. The rank prescribed by Theorem 5.2(b) for $n = 4$ is three, proving the

contradiction. The group has order 32 and exponent four, but neither a small-group classification nor a separate order bound is used in this argument. □

The existence portions of Theorem 5.2(b),(c) and the reversal in (a) are not refuted by these examples. Proposition 5 addresses the additional universal nonlinear choice, and Proposition 4 addresses the assertion of an embedding for every outer reversal.

4 Two distinct problems with the preliminary parabolic lemma

The source defines $G^i = G_{I\setminus\{i\}}$ and $G^J = \langle G^i : i \in J\rangle$, so $G^\varnothing = 1$. Lemma 2.4(c),(d) on pp. 4–5 assert the complementary equalities between G_J and $G^{I\setminus J}$ for every subset, including the extremes. At $J = I$ in (c), or $J = \varnothing$ in (d), this forces $G_I = 1$, which residual connectedness does not force for an unfaithful coset model.

Proposition 6. *Under the literal conventions of Lemma 2.4, residual connectedness does not imply (c)–(e). Conversely, (e) alone need not imply residual connectedness, even for a thin flag-transitive coset geometry with trivial Borel subgroup.*

Proof. For the first assertion take $G = \langle x, y, z\rangle \cong C_2^3$ and $G_0 = \langle y, z\rangle$, $G_1 = \langle x, z\rangle$. The coset geometry is $K_{2,2}$, so it is thin, flag-transitive and residually connected. Its Borel is $\langle z\rangle \neq 1$, disproving (c),(d) at the extremes. Moreover

$$G^{\{0\}} \cap G^{\{1\}} = G_1 \cap G_0 = \langle z\rangle \neq G^\varnothing,$$

so (e) also fails.

For the converse use the same G, but $G_0 = \langle x\rangle$ and $G_1 = \langle y\rangle$. The four generated-support groups are $1, \langle y\rangle, \langle x\rangle, \langle x, y\rangle$ and satisfy all intersection identities in (e). Their union generates only $\langle x, y\rangle < G$. The coset geometry consists of two disjoint copies of $K_{2,2}$, one in each coset of this subgroup, so it is not residually connected. It is nevertheless thin and flag-transitive: every incident pair has a common representative and is a translate of the base chamber. Here the Borel is trivial. Thus triviality of the Borel alone does not repair the claimed equivalence; the generation issue is independent. □

These examples concern the additional equivalences in Lemma 2.4, not the separate residual-generation criterion in Lemma 2.3. They do not show that every subsequent use fails; such a use must be checked with its actual generation and faithfulness hypotheses.

5 An outer automorphism quotient is not an action

The opening of Section 5.2 (p. 18) takes $B = \mathrm{Out}(A)$ and says that it naturally acts on A. The quotient $\mathrm{Aut}(A)/\mathrm{Inn}(A)$ does not in general act on the elements of A by choosing representatives: representatives of one outer class differ by an inner automorphism. To obtain the intended action inducing those outer classes, one needs a homomorphic section of

$$1 \longrightarrow \mathrm{Inn}(A) \longrightarrow \mathrm{Aut}(A) \longrightarrow \mathrm{Out}(A) \longrightarrow 1.$$

This is more than a lack of a canonical choice.

Proposition 7. *For the dihedral group $A = \langle r, s \mid r^8 = s^2 = 1,\ srs = r^{-1}\rangle$ of order 16, the map $\mathrm{Aut}(A) \to \mathrm{Out}(A)$ has no homomorphic section.*

Proof. The elements of order eight generate the unique cyclic subgroup $\langle r\rangle$, so every automorphism is uniquely

$$f_{a,b} : r \mapsto r^a, \quad s \mapsto r^b s, \qquad a \in \{1, 3, 5, 7\}, \quad b \in \mathbb{Z}/8.$$

All these maps are automorphisms. Composition is $f_{a,b}f_{c,d} = f_{ac,b+ad}$, with parameters modulo eight. The inner automorphisms have $a \in \{1, 7\}$ and even b; they are generated by $f_{7,0}$ and $f_{1,2}$. Thus $\mathrm{Out}(A) \cong C_2^2$, with generating classes $X = [f_{1,1}]$ and $Y = [f_{3,0}]$.

An involutory lift of X has the form $f_{7,c}$ with c odd: the other possibility $f_{1,c}$ has square $f_{1,2c} \neq 1$. An involutory lift of Y is either $f_{3,b}$ with b even, or $f_{5,b}$ with $b \in \{0, 4\}$. Commuting $f_{7,c}$ with the first kind would require $c + 7b = b + 3c$, or $c - 3b = 0 \pmod 4$, impossible by parity. Commuting it with the second kind would require $4c - 6b = 0 \pmod 8$, also impossible for odd c and $b = 0, 4$. A section would send the two commuting involutions X, Y to commuting involutory lifts, so no section exists. □

A chosen subgroup of actual automorphisms, such as the particular reversal $\langle\tau\rangle$ in Theorem 5.2, does define an action; this objection does not apply to that specified action. Nor does it deny the existence of unrelated homomorphisms from $\mathrm{Out}(A)$ to $\mathrm{Aut}(A)$, such as the trivial homomorphism. The missing interface is an action lifting the quotient classes.

6 Evidence and review boundary

The accompanying standard-library program `finite_models.py` constructs left cosets and group operations independently of the author's GAP script. It checks the 48-element wreath model, every coordinate support intersection and partial-flag extension for ranks three and four, all triples of distinct involutions in the two reversal extensions, both rank-two parabolic examples, and all involutory lifts used in Proposition 7. The full arguments above establish the uniform claims; the finite program also makes each particular countermodel reproducible. No Lean formalization or whole-source recovery is claimed. The three main proposed obstructions originated in the current N006 recovery work; this independent review verifies their source hypotheses and supplies the direct involution proof and the additional preliminary countermodels. No historical-priority claim is made.

References

[1] C. A. Piedade and P. Tranchida, *The Geometry of Wreath and Semi-Direct Products*, arXiv:2605.04646v2, submitted 28 May 2026; dated 29 May 2026, 22 pages.